EPSR-test-ex50 partials: all 21 site-site g(r), chained accumulation

Program(s): partials | chained passes: 2 | verification: ALL IDENTICAL (all 2 passes)

What this case tests

Exercises `partials`, the EPSR analysis program [1] that computes the intermolecular site-site radial distribution functions g(r) of every pair of atom types of an `.ato` configuration and averages them over successive configurations (`nsumt`). This is how an EPSR user obtains the partial g(r) of the refined model [2].

Code exercised

Fortran (original, fortran_src/):

- `partials.f`: `run_partials` (keywords, normalisation, pair list), `find_partials` (minimum-image pair histograms)
- `read_and_save_files.f`: `read_input_file`, `initgofz`, `outgofz` (histogram file, running average and spread), `save_files_others`; `auxiliary_routines.f`: `r_int`
- `setup_symbols_generic.f`: `setup_symbols_partials` (keyword table); `ato_read_and_save.f`: `normalise_coordinates`

Rust (port, rust/epsr-core/src/):

- `tools/partials.rs`: `run`, `find_partials`
- `tools/anal_common.rs`: `outgofz`, `write_fort14`
- `symbols.rs`: `setup_input_file`, `read_input_file`, `save_inp_file`; `output.rs`: `r_int`; `symtab_tools.rs`: `PARTIALS`

Physical system

Analysed configuration: `200DMSO400Water.ato`, an unchanged copy of the EPSR-test box of 200 dimethyl sulphoxide (DMSO; atom types S, O, C, H) and 400 water molecules (O1, H1): 3200 atoms, DMSO mole fraction 1/3, 300 K, cubic box of side 33.1213 Å (0.0881 atoms/Å^3), the EPSR model of the deuterated DMSO-water neutron data (ISIS SANDALS runs SLS03863, SLS03871, SLS03850), cf. [3]. With no centre atom (`atom-c`, `atom-z`, `atom-r` all `*`) the program histograms all intermolecular distances of each of the 6·7/2 = 21 type pairs and divides by the ideal-gas number in each shell, giving g(r) [4,5] on 300 points out to half the box side (`rnormalise 1.0`; the reference r grid ends at 16.56 Å).

Inputs

`all.PARTIALS.dat` written by hand from the defaults printed by the Fortran program: `nz 300`, `rnormalise 1.0`, `nsumt 0`, `atom-c/z/r *`, `ndist 1` (recomputed by the program); `200DMSO400Water.ato` copied unchanged from EPSR-test.

How the accuracy was verified

`commands.txt` (1 program call, repeated 2 times, each pass reading the files written by the previous one) was run with the Fortran reference build (`rust/tools/build_fortran_reference.sh`: the original sources, gfortran -O2 -fno-tree-vectorize, single thread) and with the Rust `epsr-rs` on identical copies of the inputs, and every file written was compared byte for byte (`rust/tools/run_ex_test.sh`, `compare_outputs.py`; no masking except where noted below). The Fortran output is stored in `fortran_reference.tar.gz`. Result of the final regression run of the complete port: every file identical in all 2 passes.
`repeat 2`: pass 1 starts the accumulation (`nsumt 0`), pass 2 reads `all.PARTIALS.g01` back through `r_int` and forms the running average and spread in `outgofz`. Compared after each pass: `all.PARTIALS.g01` (21 g(r) with their spreads), the rewritten `all.PARTIALS.dat` (the Fortran reference after pass 2 has `nsumt 2`, `ndist 21`) and `fort.14`, the list of the 21 pairs S-S ... H1-H1. The configuration is the same in every pass, so the chained passes test the read-back of the accumulated file and the running-average arithmetic.

Notes

`fort.14`: the Fortran writes these lines to unit 14 after `save_files_others` has closed it, so gfortran creates the default file `fort.14`; the port writes the same file.

References

[1] A. K. Soper, "Empirical Potential Structure Refinement - EPSRshell: a user's guide", Technical Report RAL-TR-2011-012, STFC Rutherford Appleton Laboratory (2011).
[2] A. K. Soper, "Partial structure factors from disordered materials diffraction data: An approach using empirical potential structure refinement", Phys. Rev. B 72, 104204 (2005).
[3] A. K. Soper and A. Luzar, "A neutron diffraction study of dimethyl sulphoxide-water mixtures", J. Chem. Phys. 97, 1320-1331 (1992).
[4] J.-P. Hansen and I. R. McDonald, Theory of Simple Liquids, 3rd ed., Academic Press (2006).
[5] M. P. Allen and D. J. Tildesley, Computer Simulation of Liquids, Oxford University Press (1987).
[6] GNU Fortran (gfortran) 15.2, GCC, https://gcc.gnu.org/fortran/ - reference build with -O2 -fno-tree-vectorize.